ORION

AI Business Operating System

Input → Processing → Output
Specification Document

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1. INPUT — What the User Provides

User's Perspective

The user provides exactly ONE thing: a single sentence.

```
"Ice cream shop in Andheri West, budget ₹15 lakh"
```

No forms. No dropdowns. No multi-page questionnaires.
One text field, one sentence — the system does the rest.

What the System Extracts

- Business Type → Ice cream shop
- Location → Andheri West, Mumbai
- Budget → ₹15,00,000
- Sector → Food & Beverage
- Scale → Small (₹10L–50L)
- Is Physical? → Yes (Operations dept will run)

Input Requirements

- Length: 10–1000 characters
- Language: English (supports Hindi-English mix)
- Format: Free-form natural language
- Required: Business type only; rest is inferred

What the User Does NOT Need

- [X] Market research data
- [X] Financial projections
- [X] Competitor names
- [X] Branding preferences
- [X] Marketing channel knowledge
- [X] Operations expertise

The system produces ALL of the above autonomously.

2. PROCESSING — What Happens Inside

Pipeline (60–90 seconds total)

- Stage 1: Intent Classification (~2s)
  `LLM extracts structured JSON from the text`
- Stage 2: Routing Decision (<100ms)
  `Decides which departments run, in what order`
- Stage 3: 21 Agents Execute (40–70s)
  `Strategy -> Finance | Branding | Marketing -> Operations`
- Stage 4: Output Assembly (~3s)
  `JSON → HTML Report → PDF`

The 21 Agents Across 5 Departments

- Strategy (5 agents): Market Sizing, Competitors,
  `Positioning, SWOT, Competitive Advantage`
- Finance (4 agents): Cost Structure, Revenue Model,
  `P&L Forecast, Break-even Analysis`
- Branding (4 agents): Identity, Voice, Visual Direction,
  `Brand Story`
- Marketing (4 agents): Digital Plan, Buyer Personas,
  `Campaign Calendar, SEO & Content`
- Operations (4 agents, physical only): Supply Chain,
  `Staffing, Technology, Process Flow`

External Data (Not Hallucinated)

- Google Places / OSM → Real competitors with ratings
- Serper.dev → Market research, industry data
- ChromaDB → Cached results for consistency

Quality Controls

- Every LLM response validated by Pydantic schema
- Revenue cannot exceed market size (enforced in code)
- Setup costs cannot exceed stated budget
- Provider fallback: Groq → Gemini → Ollama
- Append-only state: agents cannot overwrite each other

3. OUTPUT — What the System Delivers

Output Formats

- Interactive Web Report (browser, department tabs)
- PDF Document (branded, cover + TOC + page numbers)
- JSON Package (machine-readable, API accessible)

Complete Plan Contents

- Market Sizing: TAM / SAM / SOM in ₹
- Competitors: Real businesses from Google Maps
- Setup Costs: Category-wise breakdown
- Monthly Costs: Rent, staff, materials, marketing
- P&L Forecast: 20-period with margins
- Break-even: Timeline and assumptions
- Brand: Name, tagline, voice, colours, visual style
- Marketing: Personas, channels, 90-day campaign
- Operations: Staff, supply chain, daily workflow
- Transparency: Model, API, time metadata

Target Users

- First-time Entrepreneur → actionable blueprint
- Student / Researcher → structured case study
- Small Business Owner → market validation
- Freelancer / Consultant → 20–40 hours saved
- Investor / Mentor → quick viability check

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vs Generic LLM (ChatGPT etc.)

- Structure: 26-field rubric vs prose paragraphs
- Numbers: Sector benchmarks vs made-up figures

- Competitors: Real (Google Maps) vs hallucinated
- Finance: 20-period P&L vs single estimate
- Consistency: Cross-dept reconciliation in code
- Transparency: Full provenance vs none